

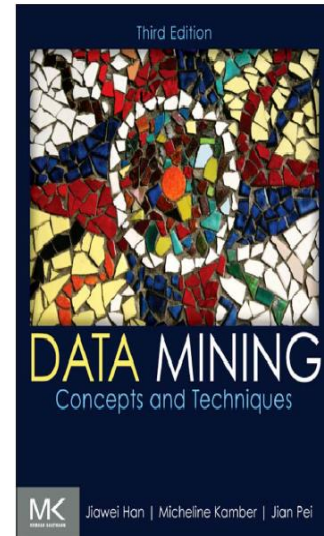
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Data Mining & Business Intelligence Course_IT674

Data Mining

Business Intelligence Reporting and Analysis

Text Book: “Data Mining: Concepts and Techniques”, Third Edition



Outlines

- Business Intelligence Overview.
- Business Intelligence Reporting.
- Business Intelligence Analysis.

Business Intelligence: Overview

■ Business Intelligence

- A set of methods, processes, architectures, applications, and technologies that gather and transform **raw data** into **meaningful and useful information** used to enable more effective strategic, tactical, and operational **insights** and **decision-making**.
- Utilizing different techniques to understand historical, live, and predictive data.
- The business intelligence platform comprises of many **BI reporting tools, business analytics, data visualization, data analytics**, etc.

BI: General Process

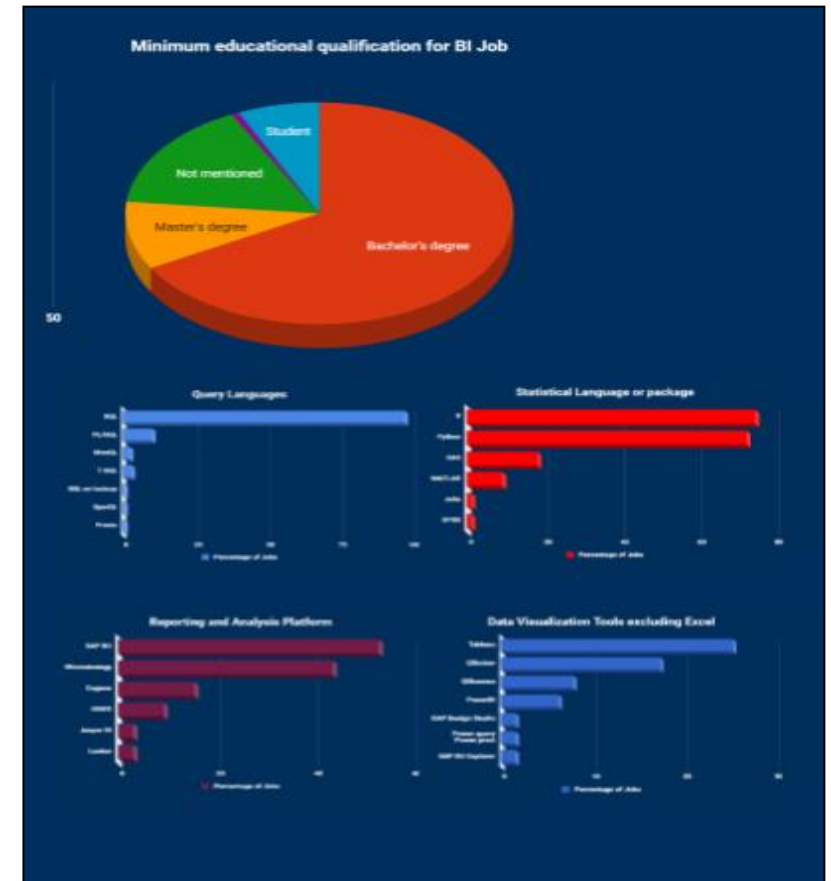


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Business Intelligence Reporting

- **Reporting**
- A foundational part of business intelligence which focuses on visualizing data in different types of visualizations such as **tables, graphs, and charts** so that data can be analyzed.
- It presents data in an interactive fashion to extract relevant insights. Ultimately, it provides suggestions and observations about business trends, empowering decision-makers to act.



The Importance of Reporting in Business Intelligence

- Reporting is one of the most important aspects of business intelligence.
- It allows businesses to track their performance and identify areas where they can improve.
- By analyzing data and generating reports, businesses can gain valuable insights into their operations, customers, and competitors.
- This information can then be used to make informed decisions that drive growth and profitability.

BI Reporting Category

- BI Reporting is divided into two categories in business intelligence:
 1. **Managed reporting:** using by technical employee such as an IT associate or data analyst prepares the data for non-technical users.
 2. **Ad-hoc reporting:** in a BI platform allows *non-technical users* to create reports from scratch or edit pre-existing reports without having to make requests from IT.

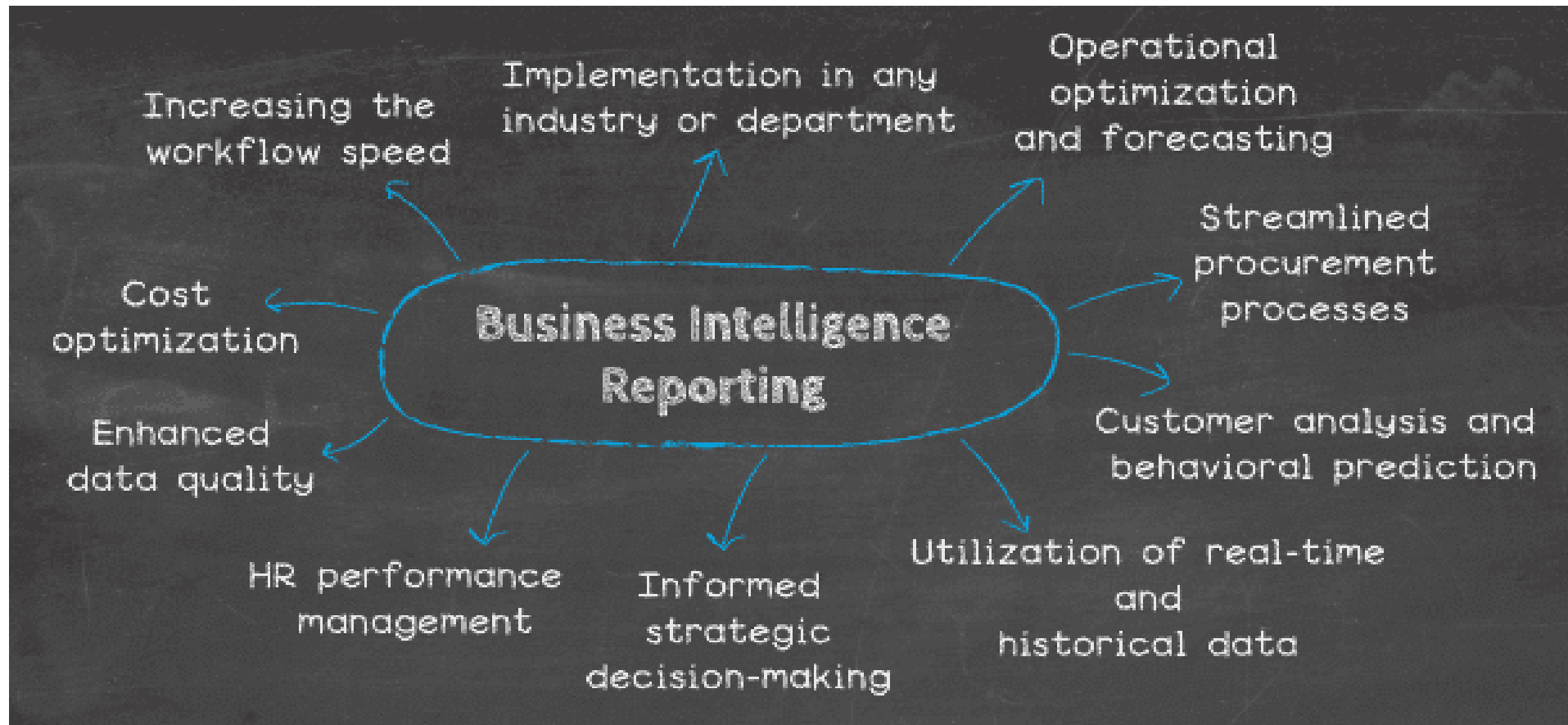
BI Reporting Features

1. BI reports can take various formats: tables, spreadsheets, and PDFs are all common outputs and can be send via email, or upload to FTP.
2. BI reports can be automated, and run on a pre-determined schedule feature which allows teams to run and send reports without manual creation or intervention.
3. **Report bursting:** where a single report can be sent to multiple users with different levels of data permission allowing them to receive a report customized to the data they need. **This saves the time and energy.**
4. BI reports can also be customized dashboards accessed through a web browser.

The Goal of Reporting in BI

- ✓ provide data insights to support decision making;
- ✓ make data easy to understand so that accurate analysis can be efficiently carried out;
- ✓ collect and present data to be analyzed, including historical data that can be tracked over time;
- ✓ empower end-users with the knowledge to become experts in their area of business;
- ✓ have the underlying figures to back up actions and explain decisions;
- ✓ Reporting is a necessary prerequisite of the analysis stage.

The Benefit of Reporting in BI



Designing Effective Business Intelligence Reports

Clear and concise data visualization:

Charts, graphs, and tables should be used to present data in a way that is visually appealing and easy to interpret.

Context Data :

should be presented in a way that provides context and meaning to the end-user. This can be achieved by including relevant benchmarks, historical data, and industry comparisons.

The end-user's workflow in mind:

Reports should be easily accessible and integrated into existing workflows to ensure they are used effectively.

Common Challenges in Business Intelligence Reporting

Data quality:

- Inaccurate or incomplete data can lead to incorrect conclusions and poor decision-making.
- For example, if a report on customer behavior is based on incomplete data, it may not accurately reflect their actual behavior.
- To overcome this challenge, businesses must ensure that their data is accurate and complete before generating reports.

Complexity of the data:

- To address this challenge, businesses must design reports that are easy to read and understand.
- This can be achieved by using clear visualizations and organizing data in a logical manner.

Why Data Visualization Matters



Clarity

Data visualization helps to clarify complex data sets and makes it easier to understand trends and patterns.



Efficiency

Data visualization allows for quick and efficient analysis and decision making by presenting information in a clear and concise manner.



Engagement

Data visualization can engage and captivate an audience, making it easier to communicate insights and findings.

Types of Data Visualization



Bar Charts

- display data as rectangular bars.
- They are commonly used to compare data across categories or to track changes over time.



Line Charts

- display data as a series of points connected by lines.
- They are commonly used to show trends over time or to compare data across categories.



Pie Charts

- display data as a circle divided into slices.
- They are commonly used to show the proportion of different categories in a dataset.

Best Practices

Keep it Simple

Avoid clutter and complexity in your visualizations. Use simple and clear designs to communicate your message effectively.

Use Color Effectively

Color is a powerful tool in data visualization. Use it effectively to highlight important data points and create visual interest. However, be careful not to use too many colors, which can be overwhelming and confusing.

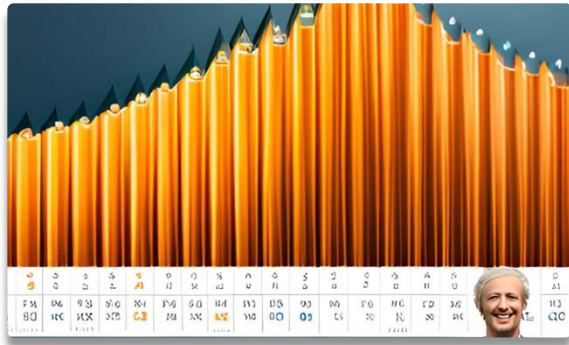
Choose the Right Chart Type

Different types of data require different types of charts. Choose the appropriate chart type that best represents your data and helps you tell your story.

Provide Context

Data visualizations should always provide context to help the audience understand the significance of the data. Use labels, titles, and annotations to provide context and tell a story.

Tools and Technologies



1. Tableau

- ease of use and
- ability to handle large amounts of data.



2. Power BI

- With Power BI, you can create interactive dashboards and reports that can help you gain insights into your business data.



3. QlikView

- known for its ability to handle large amounts of data and its powerful data analysis capabilities.

Case Studies

Airbnb

Source: Tableau

Airbnb used data visualization to identify the most popular neighborhoods in Paris and help hosts optimize their listings for maximum bookings. They used Tableau to create interactive maps and charts that allowed hosts to see how their properties compared to others in the same area and make data-driven decisions about pricing and amenities.

Key Takeaway: Data visualization can help businesses make more informed decisions and improve their bottom line.

Netflix

Source: Domo

Netflix uses data visualization to track user behavior and inform their content strategy. They use Domo to create real-time dashboards that show which shows are being watched, how long viewers are watching, and what types of content are most popular in different regions. This helps them make data-driven decisions about which shows to renew, which to cancel, and which new shows to produce.

Key Takeaway: Data visualization can help businesses stay ahead of trends and make strategic decisions about product development.

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Business Intelligence Analysis

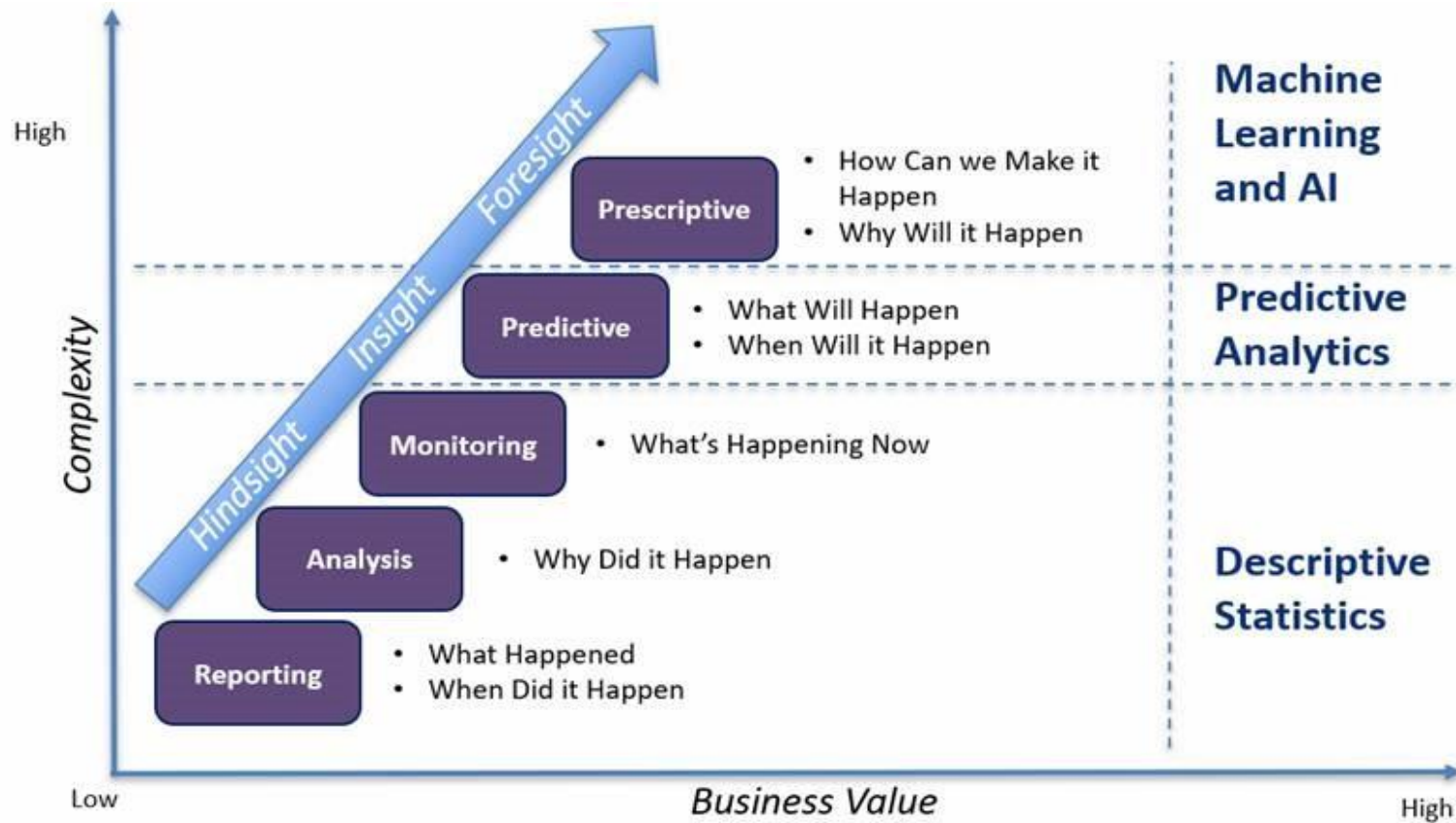
- **Analytics** refers to a more systematical, automated, and flexible **process of data analysis** for revealing insights and decision support in more extensive application areas, e.g. sports, disease, network traffic, etc.

“Analytics refers to the entire domain of leveraging information to make smarter decisions. In other words, **reporting** and **analysis**. “

BI - Analysis Tools

- **Basic querying and reporting - “Tell me what happened.”**
 - Structured and fixed format reports Based on simple and direct queries
 - Usually involves simple descriptive analysis and transformation of data, such as calculating, sorting, filtering, grouping, and formatting
- **Ad hoc query and reporting - “Tell me what happened when they need.”**
 - Similar to operational reporting but on a need basis
- **Business analysis(OLAP) - “Tell me what happened with why.”**
 - A multi-dimensional analysis and reporting application for aggregated data
 - Great for discovering details from large quantities of data
 - Business analytics (BA) is the practice of iterative, methodical exploration of an organization’s data with emphasis on statistical analysis.
- **Data mining - “Tell me what might happen” or “Tell me something interesting.”**
 - Data mining techniques are a blend of statistics and mathematics, and artificial intelligence and machine-learning.

Analytics in BI



Summary

- Business intelligence is defined as the process of taking large amounts of data, analyzing that data, and presenting a high-level set of reports that condense the essence of that data into the basis of business actions, enabling management to make fundamental daily business decisions (Stackowiak et al. 2007);
- BI includes several software for Extraction, Transformation and Loading (ETL), data warehousing, database query and reporting, multidimensional/on-line analytical processing (OLAP) data analysis, data mining and visualization;
- This lecture explores the concepts of BI Reporting and Analysis, benefits of BI, factors influencing BI, technology requirements.

Thank you

